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Research progress in online monitoring of suspended sediment concentration

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Suspended sediment concentration and transport processes are important controls on channel morphodynamics, hydraulic-engineering safety, and aquatic ecological and biogeochemical processes. Continuous monitoring supports flood regulation, reservoir operation, and navigation maintenance, and is also directly related to particulate carbon, nutrient, and contaminant transport, light attenuation, primary productivity, habitat quality, and the evolution of estuarine turbidity maxima. This framing expands SSC monitoring from a hydraulic measurement variable to a biogeoscience observation variable. This technical review uses a thematic literature search and synthesis to examine online and *in situ* monitoring methods for suspended sediment, including optical, acoustic, differential-pressure, vibration, electrical, nuclear, and sampling-based approaches. It compares their measurement mechanisms, applicable ranges, calibration needs, field deployment, and uncertainty sources. The review shows that sensor outputs cannot be directly equated with suspended sediment concentration. Site-specific conversion relations must be established by considering particle size, mineral composition, concentration range, hydrodynamic conditions, and paired sampling. Relations between turbidity, acoustic backscatter intensity, remote-sensing reflectance, and SSC are strongly site dependent, and nominal instrument accuracy cannot be directly converted into SSC accuracy. Future research should build a reproducible technical framework that integrates acoustic-optical retrieval, representative cross-sectional deployment, joint laboratory and field calibration, automated quality control, long-term drift correction, and uncertainty reporting.

KEYWORDS

acoustic–optical fusion, calibration methods, online monitoring, quality control, sensor deployment, suspended sediment

1 Introduction

Suspended sediment transport is a key control on channel morphodynamics, hydraulic-engineering safety, and aquatic ecological and biogeochemical processes. Its dynamics directly affect flood regulation, reservoir operation, navigation stability, and river-estuary ecological protection. Suspended sediment is not only the material carrier of water-sediment fluxes, but also an important medium for the transport of particulate organic carbon, nitrogen and phosphorus nutrients, heavy metals, and other contaminants (Horowitz, 2008). Changes in its concentration and particle size alter underwater light attenuation,

primary productivity, algal bloom risk, fish habitat quality, and the position and intensity of the estuarine turbidity maximum (Cloern, 1987). Continuous and accurate monitoring of suspended sediment transport therefore has hydrological, sedimentological, ecological, and biogeochemical significance (Meybeck, 1982; Owens et al., 2005; Bilotta and Brazier, 2008; Burchard et al., 2018).

With advances in sensor technology, online suspended sediment instruments based on optical, acoustic, electrical, and other principles are increasingly developed and applied. Different measurement principles now form distinct application systems in a range of hydrological settings. However, each instrument is constrained by its measurement mechanism, and its performance varies markedly with sediment concentration, hydrodynamic conditions, and particle-size distribution. A unified and widely adaptable technical solution is therefore still difficult to establish (Li et al., 2022; Wang et al., 2023; Guerrero et al., 2017; Rymaszewicz et al., 2017; Landers et al., 2016).

Existing studies provide many descriptions of measurement principles and laboratory comparisons, but integrated comparison remains insufficient for long-term operation, calibration transferability, drift correction, paired-sampling quality, and cross-sectional sediment-load uncertainty in complex natural rivers and estuaries. The key question is not only whether a signal can be detected, but under which water-sediment conditions the signal can be reliably converted into SSC, to what extent a point observation represents a vertical, cross section, or river reach, and whether the resulting error can be quantified and reported transparently. This review therefore summarizes the state of online suspended sediment monitoring technologies, identifies the applicable conditions and limitations of different principles, and clarifies the main factors and error sources that affect overall measurement accuracy. It reviews mainstream methods, field application features, deployment strategies, and accuracy assessment, with the aim of supporting high-accuracy online monitoring and engineering practice. The search scope includes CNKI, Web of Science, Scopus, Google Scholar, USGS/FISP technical reports, national and industry standards, and selected manufacturer manuals. The search terms include suspended sediment concentration, SSC, turbidity, optical backscatter, acoustic backscatter, ADCP, sediment acoustic index, remote sensing, calibration, drift, QA/QC, uncertainty, and sampler.

2 Applicability conditions of different online suspended sediment measurement techniques

2.1 Measurement principles of online sediment monitoring technologies

Suspended sediment measurement methods can be divided into direct and indirect methods according to whether the water sample is removed during measurement. Direct methods are represented by physical sampling. A sampler first collects water, and SSC is then determined by filtration and oven drying, evaporation weighing, or density-based methods. These methods provide the calibration and comparison basis for most online sensors (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015; Guy, 1970; Bureau of Hydrology,

Ministry of Water Resources, 2000). Indirect methods establish empirical or semi-theoretical relations between sensor signals and SSC to provide continuous outputs. They mainly include optical, acoustic, differential-pressure, vibration, electrical, nuclear, and integrated methods combined with automatic sampling, discharge measurement, or remote sensing. Unless otherwise stated, SSC in this paper refers to mass concentration and is expressed in mg/L or kg/m³, where 1 kg/m³ = 1,000 mg/L. NTU and FNU are turbidity units that describe scattered-light intensity and cannot be directly converted into SSC without site sediment samples, particle-size information, and regression uncertainty.

Optical measurement of suspended sediment uses a light source to emit a selected wavelength into sediment-laden water and retrieves concentration and particle-size information from scattered, transmitted, reflected, diffracted, or image-based signals (Rai and Kumar, 2015). According to signal form, optical methods include backscatter, transmission, focused beam reflectance, laser diffraction, remote sensing reflectance, and digital image analysis (Osborn et al., 2021). Transmission-based optical sensors have also been applied in sediment-concentration monitoring, but their field performance still depends on site-specific calibration, optical-path stability, particle-size composition, and water-quality interference (Zhao et al., 2022). Backscatter and transmission turbidity sensors have simple structures, relatively low cost, and high temporal resolution, but their outputs are sensitive to particle size, color, mineral composition, flocculation state, bubbles, and biofouling. They usually provide only a surrogate signal for a local point or small water volume (Baker and Lavelle, 1984; Downing, 2006; Merten et al., 2014; Zheng et al., 2017). Focused beam reflectance and laser diffraction can provide particle-size distribution information (Yu and Erickson, 2008). Laser diffraction can estimate particle size and concentration in laboratories and some *in situ* settings, but retrieval depends on refractive index, particle shape, optical path, and multiple-scattering assumptions, and is not independent of particle size (Gartner et al., 2001; Guerrero et al., 2014). Remote sensing expands spatial coverage, but its main limitation is not simply high velocity or strong turbulence. It obtains information only from the surface layer or a limited optical depth and is constrained by atmospheric correction, sun glint, surface roughness, bottom reflectance, mixed water-color constituents, synchronous field sampling, and conversion from surface SSC to depth-integrated SSC (Ritchie et al., 2003; Dethier et al., 2020; Adjovu et al., 2023). Recent optical water-quality monitoring emphasizes the integration of *in situ* point sensors, automatic stations, UAV or satellite remote sensing, and machine-learning methods, but power consumption, drift, antifouling capacity, and calibration transferability remain key limits for long-term use of low-cost sensors (Kumar et al., 2024). Literature sometimes cited in relation to quantum-dot spectral measurement actually discusses the volume scattering function of natural bubbles rather than quantum-dot SSC sensors, so this method should not be treated as a mature online sediment-monitoring approach.

Acoustic measurement emits fixed-frequency or multi-frequency sound waves and retrieves SSC from backscatter or transmission attenuation produced by sediment particles (Guerrero et al., 2011). Acoustic backscatter retrieval must account for propagation loss, water absorption, sediment absorption and scattering attenuation, particle-size distribution, particle density,

and instrument geometry. Its calculation is therefore more complex than the conversion of single-point optical turbidity (Thorne and Hanes, 2002; Sassi et al., 2012). ADCP, ABS, H-ADCP, and side-looking acoustic profilers can support velocity profiling or discharge measurement and can also serve as surrogate SSC monitoring tools after rigorous calibration. Their effective SSC retrieval range depends on frequency, blanking distance, acoustic path length, sediment concentration, particle size, attenuation, signal-to-noise ratio, and paired field sampling. Recent USGS technical guidance on acoustic sediment monitoring further stresses that deployment decisions should be made according to the monitoring objective across horizontal or vertical, point or profile, single-frequency or multi-frequency, continuous or discrete, and fixed or mobile modes, and that operational applications require calibration checks, long-term drift assessment, and uncertainty reporting (Landers et al., 2016; Wood et al., 2023; Alexander et al., 2023; Snazelle, 2017). Low-cost ABS and multi-frequency acoustic-optical fusion can improve spatial-temporal coverage and particle-size adaptability, but synchronous physical sampling, particle-size observation, and drift correction are still needed to prevent failure of empirical relations (Guerrero et al., 2017).

In addition to optical and acoustic methods, differential-pressure and vibration methods are used in limited settings. Differential-pressure methods place pressure sensors at two fixed vertical elevations to determine the density difference between two points, calculate the density of the sediment-laden fluid, and convert density into sediment concentration. These methods can measure relatively high concentrations above 10 kg/m³ and are stable, without biofouling or signal-drift problems in the optical sense. However, because they rely on an assumed constant concentration between measurement points, they are not well suited to rapidly varying SSC and are affected by pressure-sensor accuracy and flow disturbance. They are not appropriate for low-concentration measurements (Hsu and Cai, 2010). Vibration methods use the fact that fluids of different density induce different vibration frequencies in a vibrating tube or fork. Sediment concentration is estimated from the measured vibration signal through a concentration-frequency relation. These methods are suitable for high-concentration settings and are low cost, but they are strongly affected by water temperature and pressure fluctuations, which can bias the results (Hsu et al., 2008; Zhou et al., 2012).

Electrical and nuclear methods also exist (Li et al., 2005; Maucec and Denijs, 2009). Electrical methods measure SSC through relations between the apparent dielectric constant, conductivity, capacitance, or impedance response of the sediment-water mixture and SSC. They can have a wide potential range, but are strongly affected by temperature, salinity, bubbles, organic matter, algae, and electrode contamination, and their signal-to-noise ratio is low under low-concentration conditions. Nuclear methods estimate sediment concentration from changes in gamma-ray or X-ray transmission, scattering, or absorption. They can operate in high-turbidity or optically opaque media, but long-term hydrological sediment monitoring is limited by radioactive-source safety, regulatory approval, licensing, transport and maintenance requirements, and field operating cost. Their application range is therefore limited. Table 1 compares the principles, applicable conditions, calibration requirements, and limitations of these technologies.

To strengthen the integrated description of different measurement technologies, this review summarizes online and *in situ* suspended sediment monitoring as a chain from method family to raw signal, field conversion, and SSC product (Figure 1). The figure emphasizes that online instrument outputs, such as turbidity, voltage, acoustic echo, density response, and reflectance, cannot be directly equated with SSC. Reliable concentration estimates must rely on field sampling, particle-size information, drift checks, quality control, and uncertainty assessment.

Based on the summary of applicable conditions, calibration needs, and limitations in Table 1 and Figure 2 further compares method suitability from the perspectives of water-sediment conditions and operational constraints. The figure does not indicate the absolute accuracy of any instrument type. Instead, it highlights the relative advantages and limitations of different technologies under low or high SSC, variable particle size, cross-sectional representativeness, spatial coverage, long-term operation, and maintenance cost.

2.2 Key influencing factors of *in situ* sediment measurement principles

Different technologies have different measurement principles and different influencing factors. Correct evaluation of these factors helps select suitable instruments in different monitoring settings and quantitatively identify the causes of anomalous measurements (Niu and Lai, 1996). Studies of influencing factors usually combine instrument experiments with theoretical analysis. Because optical and acoustic instruments are the most extensively studied, whereas other techniques have narrower application ranges, stronger empirical dependence, and in some cases no mature commercial products, this review focuses on existing work on optical and acoustic methods.

Instruments based on optical scattering and transmission are commonly called turbidity meters. Sediment concentration has the largest effect on the measured signal, with an order of influence of $O(1,000)$. Particle size has the next largest effect, $O(100)$, particle shape has a smaller effect, $O(10)$, and flocculation or disaggregation has the smallest effect, $O(2)$ (Downing, 2006). When SSC is below 6 kg/m³, the OBS optical signal increases almost linearly with concentration. Above 6 kg/m³, the linear trend weakens toward a peak, and the signal then decreases as concentration continues to rise. Because more sensitive optical receivers respond more strongly to concentration change, the critical concentration varies slightly among receiver designs. Sensors are most sensitive to fine particles, especially silt and clay, and sensitivity decreases as particle size increases. For OBS, the signal response to 200–400 μm particles is least affected by changes in particle-size distribution, whereas particles smaller than 44 μm produce the strongest sensitivity to grading changes (Conner and De Visser, 1992). OBS instruments are therefore not recommended where the particle-size distribution changes and particles are smaller than 100 μm . At the same SSC, different particle shapes can produce markedly different signals at different detection angles, although quantitative results on the effect of particle shape remain limited. Flocculation or disaggregation acts mainly by changing the effective particle size and shape. Sediment color, dissolved substances, bubbles, chemical or biological fouling, and aquatic organisms can also

TABLE 1 Summary of applicability, calibration requirements, and limitations of online suspended-sediment concentration monitoring technologies (Li et al., 2022; Rai and Kumar, 2015; Wren et al., 2000; Haddadchi, 2017; Downing, 2006).

Technology	Measurement signal/principle	Applicable conditions	Calibration requirements	Maintenance needs	Limitations and representativeness issues
Optical backscatter/transmission	Uses visible or near-infrared light; retrieves SSC from backscatter intensity, transmitted-light attenuation, turbidity, or voltage output	Low to moderate SSC; relatively stable particle size, color, mineral composition, and optical-water properties; fixed-point automatic stations	Requires site-specific calibration with physical samples across the expected SSC and particle-size range. Turbidity units such as NTU/FNU cannot be directly converted to SSC without regression and uncertainty assessment	Requires frequent cleaning of optical windows, anti-fouling measures, zero/span or blank checks, inspection after floods, and recalibration when sediment properties change	Sensitive to particle size, color, flocculation, bubbles, biofouling, and high-turbidity saturation. Usually represents only a local point or small sampling volume, so vertical or cross-sectional mean SSC must be verified separately
Focused beam reflectance	Uses a focused laser beam and reflected signals to obtain chord-length distributions of particles or flocs	Laboratory or short-term field diagnosis of particle aggregation, flocculation, and particle-size variation; auxiliary tool for interpreting concentration sensors	Requires calibration or interpretation models linking chord-length distributions to particle size or SSC, with validation using independent particle-size and SSC samples	Requires careful probe cleaning, optical-window inspection, control of alignment and immersion conditions, and checks for fouling or bubbles	High cost and local measurement volume. Results depend on particle shape, refractive properties, particle overlap, bubbles, and probe methodology; field representativeness must be evaluated
Laser diffraction	Retrieves particle-size distribution and related concentration information from multi-angle scattering or diffraction signals	Laboratory calibration, short-term in situ profiling, and diagnosis of particle-size changes where optical path and concentration are suitable	Requires assumptions or measurements of refractive index, particle shape, and optical model parameters; must be checked against physical samples, especially under changing particle-size distributions	Requires optical-path cleaning, alignment checks, protection from bubbles and fouling, and verification after high-turbidity events	Not independent of particle size. Multiple scattering and high turbidity reduce reliability; equipment is costly and long-term unattended deployment is difficult
Spectral reflectance/remote sensing	Uses satellite, UAV, or shore-based reflectance to retrieve surface SSC through empirical, index-based, or semi-analytical models	Spatial mapping in estuaries, lakes, reservoirs, river plumes, large rivers, and coastal waters; scale extension when combined with fixed stations or surveys	Requires synchronous or near-synchronous field SSC samples, atmospheric correction, model training and validation, and uncertainty reporting for surface-to-depth or surface-to-flux conversion	Requires sensor/platform radiometric checks, image quality screening, cloud and sun-glint removal, and periodic recalibration under seasonal or hydrological changes	Mainly represents the surface layer or limited optical depth. Affected by atmosphere, sun glint, surface roughness, bottom reflectance, mixed water constituents, and weak transferability among sites
Digital image analysis	Extracts particle or floc size, shape, number, and structure from high-resolution images using segmentation, recognition, and statistical algorithms	Laboratory and in situ studies of particle processes, floc structure, and development of new monitoring methods; useful where water clarity and imaging conditions are adequate	Requires calibration against physical SSC and particle-size measurements; image-processing algorithms need training, validation, and site-specific quality checks	Requires stable illumination, focus, lens/window cleaning, camera protection, storage and power management, and regular checks of algorithm performance	Particle overlap, shielding, turbidity, bubbles, fouling, and illumination variability reduce reliability. Data volume and algorithm transferability limit operational use

(Continued)

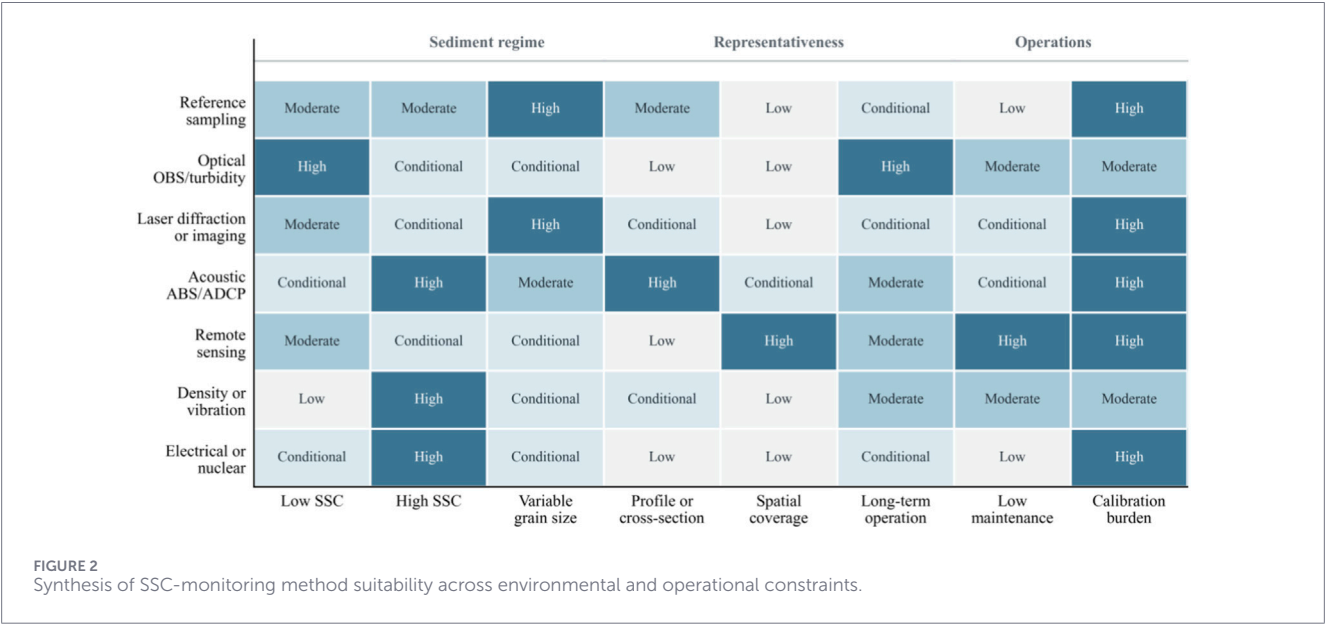
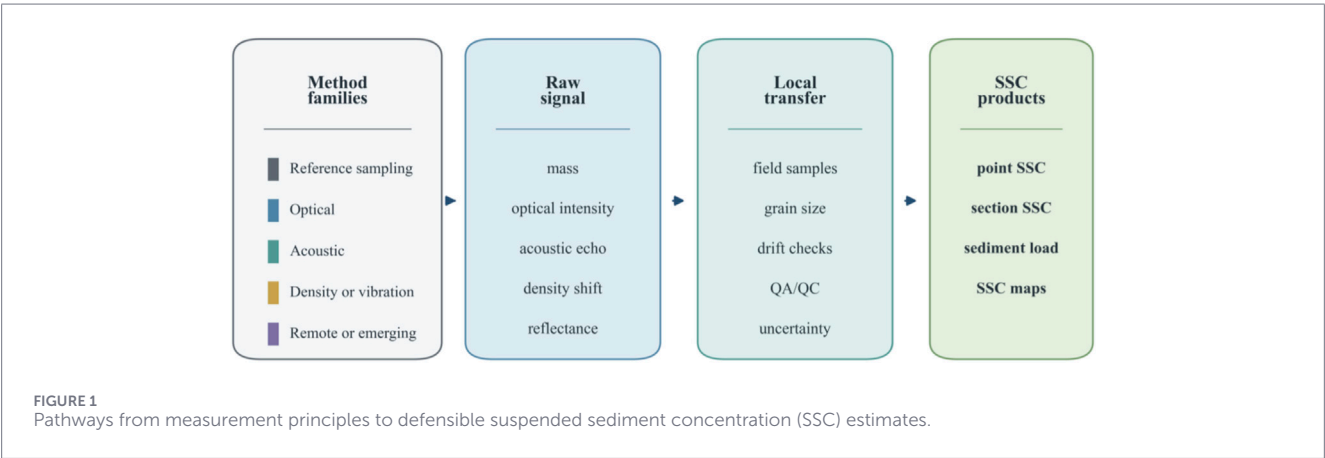
TABLE 1 Continued

Technology	Measurement signal/principle	Applicable conditions	Calibration requirements	Maintenance needs	Limitations and representativeness issues
Acoustic backscatter/transmission	Uses fixed- or multi-frequency sound waves; retrieves SSC from acoustic backscatter or transmission attenuation together with particle-size and attenuation information	Turbid or low-light waters; beam or profile monitoring; integration with ADCP, ABS, H-ADCP, or side-looking profilers for combined flow and sediment monitoring	Requires paired physical sampling, particle-size information, acoustic attenuation correction, frequency-dependent calibration, drift checks, and uncertainty assessment	Requires transducer cleaning, mounting-angle checks, inspection for biofouling and bubbles, signal-to-noise monitoring, firmware/time synchronization, and post-flood validation	Retrieval is complex and depends on frequency, particle size, mineral composition, concentration, attenuation, acoustic path length, boundary reflection, and signal-to-noise ratio. Beam/profile data still require validation for cross-sectional mean SSC
Differential-pressure principle	Estimates mixture density from pressure differences between two fixed vertical elevations and converts density to SSC	High-concentration sediment-water mixtures or engineering slurry settings where vertical density assumptions are acceptable	Requires calibration between pressure difference, mixture density, and SSC using physical samples; pressure-sensor zero drift and elevation difference must be checked	Requires pressure-port cleaning, blockage prevention, sensor drift checks, protection from impact, and verification of elevation difference	Low sensitivity at low SSC. Results are affected by turbulence, bubbles, temperature, salinity, blockage, pressure drift, and the assumption that concentration between measurement points is representative
Vibration principle	Uses changes in vibration frequency or vibration characteristics of a vibrating tube, fork, or similar structure caused by mixture density	High-concentration sediment-water mixtures, slurries, or fixed engineering facilities where optical methods may saturate.	Requires concentration-density-frequency calibration with physical samples and compensation for temperature, pressure, salinity, and installation effects	Requires cleaning to prevent scaling or deposition, inspection for wear and external vibration, temperature/pressure compensation checks, and periodic field comparison.	Intrusive installation may disturb local flow. Low-SSC resolution and relative error must be verified; results are affected by bubbles, scaling, pressure, temperature, salinity, wear, and external vibration
Electrical property method	Measures capacitance, conductivity, dielectric constant, impedance, or other electrical responses of sediment-laden water	Exploratory or site-specific monitoring where water chemistry is stable and optical transparency is poor	Requires local calibration relating electrical response to SSC while accounting for water temperature, salinity, ionic composition, and sediment properties	Requires electrode/probe cleaning, anti-corrosion measures, checks for algae or biofouling, electrical stability checks, and recalibration when water chemistry changes	Temperature, salinity, organic matter, ionic composition, bubbles, algae, and electrode contamination may mask the SSC signal; signal-to-noise ratio is weak at low concentrations
Nuclear technique	Estimates SSC from gamma-ray or X-ray transmission, scattering, or absorption changes through sediment-laden media.	Special industrial or controlled engineering environments; high-turbidity or optically opaque media where regulatory and safety conditions can be satisfied.	Requires source-detector calibration, density/SSC calibration, radiation-safety verification, licensing, and routine performance checks	Requires radioactive-source management, detector maintenance, safety training, regulatory records, shielding inspection, and emergency procedures	Limited by radiation safety, licensing, transport, maintenance cost, and operator requirements; generally unsuitable for widespread ordinary hydrological-station deployment

(Continued)

TABLE 1 Continued

Technology	Measurement signal/principle	Applicable conditions	Calibration requirements	Maintenance needs	Limitations and representativeness issues
Sampling method	Collects a defined water volume manually or automatically; SSC is determined by filtration and oven drying, evaporation weighing, or laboratory analysis	Reference method for online-sensor calibration and validation; event sampling, depth-integrated sampling, and time-integrated sampling	Requires representative sampling design, isokinetic or validated intake conditions, laboratory QA/QC, repeated samples, and matching between sample time and sensor signal	Requires sampler cleaning, intake and bottle inspection, preservation of samples, laboratory quality control, and maintenance of sampling logs	Low temporal resolution and high labor demand. Non-isokinetic sampling, point representativeness, sample handling, and laboratory errors can propagate into sensor calibration and sediment-load estimates



affect measurements. Under otherwise constant conditions, OBS sensitivity is positively related to Munsell value, where larger values indicate lower darkness. Black sediment produces the lowest OBS gain response, white sediment produces the highest response, and other colors fall between them, because darker particles absorb near-infrared light and alter the output. This implies that, for the same SSC, darker sediment may yield a lower estimated concentration than lighter sediment (Sutherland et al., 2000). Light-absorbing

dissolved organic or inorganic matter can reduce scattered-light intensity and may cause measured SSC to be lower than the true value. Whether bubble effects can be ignored remains an active research question. Existing studies suggest that bubbles mainly affect backscattering efficiency. Small bubbles of 30–90 μm have an effect about one order of magnitude smaller than sediment particles, whereas larger bubbles have greater weight. Accounting for the particle-size effects of both bubbles and sediment is the most effective way to reduce bubble interference (Zhang et al., 2002). Chemical and biological fouling and organisms can interfere with light emission and reception and may generate stray scattering, leading to either positive or negative bias.

Acoustic measurement develops later than optical backscatter and transmission, and sediment concentration and particle size are now recognized as the main factors controlling acoustic results. Experiments show that as SSC increases, the relation between received acoustic signal and SSC becomes progressively nonlinear, and acoustic inversion becomes less applicable. Instrument response depends on particle size. For fixed instruments, the emitted acoustic frequency is fixed, commonly between 200 kHz and 3 MHz, so the applicable particle-size range is also constrained. For a given particle-size distribution, an optimal acoustic frequency exists for concentration measurement. Higher frequencies are more suitable for smaller particles, whereas lower frequencies are more suitable for larger particles. At the same SSC, measurement error increases as the particle-size distribution changes. Most instruments therefore use two or three acoustic frequencies to reduce particle-size interference. As in optical methods, differences in mineral composition and density strongly affect acoustic SSC measurement, and assuming uniform quartz particles can produce substantial error (Moate and Thorne, 2012). Dissolved substances, bubbles, fouling, and organisms in water are also expected to affect acoustic measurements, but fewer studies quantify these effects.

2.3 Reported applicability conditions and accuracy ranges of existing instruments

Instruments based on optical backscatter, transmission, and diffraction usually specify nominal working ranges for temperature, water depth, pressure resistance, turbidity, or concentration, but these parameters are not equivalent to SSC retrieval accuracy. For turbidity sensors, indicators such as $\pm 1\%$ to $\pm 5\%$ or 0.01 to 0.5 NTU mainly describe turbidity or electrical-signal performance and cannot be directly converted into SSC accuracy. SSC accuracy also depends on site sediment particle size, color, mineral composition, flocculation state, regression model form, sample number, and calibration residuals (Rymaszewicz et al., 2017; Wang et al., 2020). Acoustic instruments also require a distinction between velocity-profile capability and SSC retrieval capability. ADCPs can measure velocity profiles in deep water, but their effective acoustic path and concentration range for SSC retrieval are jointly controlled by frequency, blanking distance, acoustic attenuation, particle scattering efficiency, signal-to-noise ratio, and synchronous sampling calibration. Multi-frequency acoustics can reduce particle-size dependence, but field applications still need calibration checks, drift correction, and uncertainty ranges.

To compare applicable intervals, this review compiles particle-size and concentration information for optical, acoustic, and vibration instruments from accessible manufacturer materials and literature (Gentile et al., 2010; Merten et al., 2014; Wang et al., 2020; Rymaszewicz et al., 2017; Druine et al., 2018; Ochiai and Kashiwaya, 2010; Gartner et al., 2001; Moate and Thorne, 2012; Fang and Tang, 1990; Emerson, 2024). Some particle-size ranges in Figure 3 are interpretive summaries based on Mie-scattering sensitivity, experimental boundaries in the literature, and available product parameters rather than official manufacturer specifications. Overall, optical backscatter and transmission are more suitable for low to moderate SSC and relatively stable particle-size conditions. Laser diffraction provides particle-size information but depends on optical assumptions. Acoustic methods extend monitoring to profiles or more turbid settings, but retrieval is more sensitive to frequency, attenuation, and particle size. Vibration density meters can cover higher concentrations, but sensitivity at low SSC and field maintenance remain major issues.

The LISST series can be classified as optical diffraction instruments; OBS/AQUAlogger instruments can be classified as optical backscatter/transmission instruments; AQUAscat/ADCP instruments can be classified as acoustic backscatter instruments; and Micro Motion FDM can be classified as a vibration/density-based instrument. Where manufacturers do not provide particle-size ranges, the ranges shown are interpretive judgments based on Mie-scattering sensitivity and experimental upper and lower limits, rather than official specifications (Mie, 1908; Emerson, 2024). Scatter points indicate measurement data used for calibration or reported particle-size distributions in the literature. For a given grading distribution, the median particle size is used as the representative value, and error bars indicate the maximum-minimum range reported in the same study. Acoustic scatter points include selected public instrument information. The yellow band indicates the more credible concentration interval for vibration instruments reported in the literature and manufacturer materials, and its horizontal width does not indicate particle-size dependence. The vertical axis uses mg/L.

3 Sediment *in situ* monitoring deployment methods

3.1 Selection of monitoring sections and verticals for sediment *in situ* observation

River cross sections selected for online sediment monitoring usually rely on existing long-term hydrological monitoring sections and follow the core principles of the Code for Measurement of Suspended Sediment in Open Channels (Hydrology Bureau of the Changjiang Water Resources Commission, 2023; Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015). The code requires measurement sections to be located in straight and stable reaches with limited main-flow migration and away from abnormal flow zones such as recirculation and bank collapse, so that data are representative. Online monitoring also has specific requirements because of sensor characteristics and real-time operation.

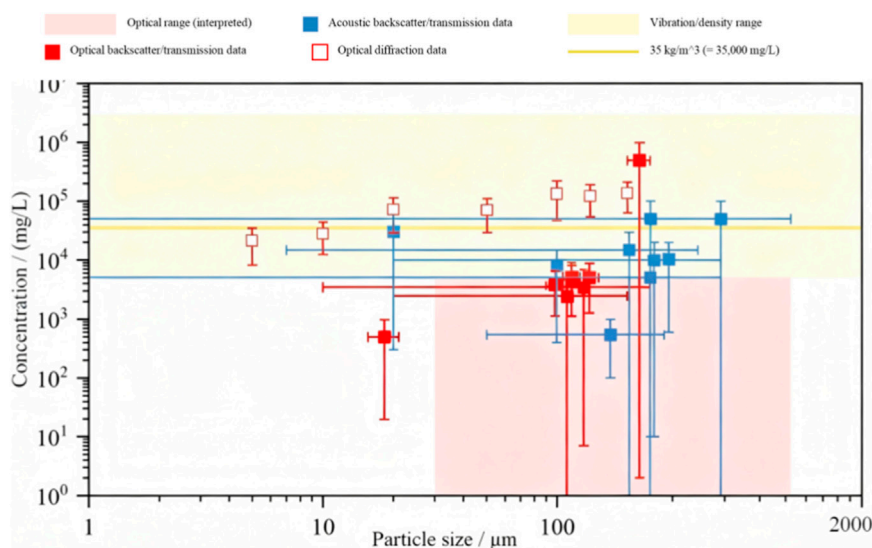


FIGURE 3
Measurable particle-size and concentration ranges of optical, acoustic, and vibration concentration-measuring instruments.

Online monitoring depends on continuous sensor observations, but it is not necessarily equal to cross-sectional mean SSC or sediment load. In addition to avoiding recirculation, bank collapse, local scour and deposition, and other abnormal zones, deployment should be selected according to the monitoring objective. Bank-based point sensors are easy to maintain but have limited spatial representativeness. Bridge or cableway systems facilitate fixed-point lifting and paired sampling. Moving-boat surveys or mobile ADCP measurements provide lateral and vertical distributions but lack continuity. Vertical arrays can characterize concentration gradients but require higher investment and maintenance. H-ADCP or side-looking acoustic profiling is suitable for integration with discharge monitoring. Pumped sampling and automatic samplers provide physical samples for sensor drift checks and event calibration. Engineering cases are useful as experience for specific reaches, but should not be directly generalized into universal deployment rules. (Le Coz J et al., 2008, Vougioukas et al., 2011).

For vertical selection, traditional standards emphasize verticals based on turning points in unit-width sediment discharge, equal partial discharge, or equal partial area to cover concentration variations across the section (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015). Online monitoring is constrained by long-term fixed installation, power supply, communication, maintenance, and safety, so it usually requires a balance between representativeness and operability (Du, 2021). One or two representative verticals are acceptable only where the reach is straight, mixing is sufficient, lateral SSC gradients are small, the target uncertainty is broad, and validation has been made with multi-vertical comparisons or historical data. For wide shallow channels, tributary confluences, bends, tidal estuaries, rapidly rising or falling floods, and reaches with complex lateral SSC distributions, additional verticals, mobile surveys, or correction using acoustic or remote sensing spatial information is required (McSweeney et al., 2016).

3.2 Selection of vertical measurement points for sediment *in situ* monitoring

Traditional manual sampling, as the classical monitoring mode, uses multi-point stratified sampling and discharge-weighted calculation as core principles (Ministry of Housing and Urban-Rural Development of the People's Republic of China, 2015). In large rivers with greater depth and width, three-point, five-point, or six-point vertical methods are commonly used. This mode measures velocity and SSC at each point and calculates vertical mean SSC through stratified weighting. It can adapt to complex flow conditions, but it has low efficiency and limited temporal and spatial resolution (Wang, 2015).

From the perspective of vertical water-sediment distribution, the central part of relative depth is commonly located in the more uniform core of the Rouse-type suspended sediment distribution. It can reduce interference from surface bubbles, near-bed bed-material effects, and abrupt vertical gradients, and is therefore often the preferred depth for an online representative point. In low-sediment rivers, however, vertical SSC distribution is more uniform, and sensor deployment and maintenance convenience often lead to relatively shallow installation. At the Lueyang hydrological station, the sensor of the TES-71 cableway sediment monitoring system is fixed 0.6 m below the water surface, corresponding to about 0.1 h for a 10 m water depth (Zhao and Xia, 2021; Wang et al., 2022). A single fixed vertical point can achieve relatively high calibration accuracy for cross-sectional mean SSC in that setting (Pan, 2021). At the Zhicheng station on the Yangtze River, the TES-91 online sediment sensor is installed at a fixed location at the stern of a hydrological pontoon, with an immersion depth of 1.85 m, or about 0.2 h (Hu et al., 2020). For special water-sediment environments, representative-point depth must be adjusted. In high-concentration flows or fluid-mud conditions, vertical SSC gradients increase markedly, and a single depth cannot cover the range of variation. In the main Yangtze River, since impoundment of the Three Gorges

Reservoir, suspended sediment concentration has decreased greatly and the vertical distribution has developed a trailing pattern. The concentration 0.1 m above the bed can reach 15 times the vertical mean (He et al., 2020). Under this strongly unsaturated sediment-transport condition, a point shallower than 0.2 h is not representative of the vertical mean concentration.

Current online sediment instruments are commonly installed too shallow, often within 0.2 h of the water surface. The depth of online vertical monitoring points should conform to the vertical distribution of suspended sediment and to the water-sediment environment, instrument characteristics, and maintenance mode. In low-concentration reaches with sufficient vertical mixing and weak concentration gradients, a shallow fixed point can provide a useful cross-sectional representative value through a stable calibration relation. In high-concentration flows, bed-sediment resuspension, salt-freshwater mixing, fluid mud, or strongly unsaturated transport, a single shallow point cannot represent vertical mean SSC and should be corrected using multiple vertical points, pumped profiles, mobile measurements, or acoustic profiles.

3.3 Deployment decision framework for online monitoring

Online monitoring deployment can follow the sequence of monitoring objective, water-sediment mixing state, maintainability, calibration conditions, and uncertainty target. If the objective is warning or trend monitoring, bank-based or fixed-point turbidity or ABS sensors can be prioritized, with event sampling used to maintain the calibration relation. If the objective is cross-sectional sediment load, discharge measurement should be considered simultaneously, and deployment should prioritize representative verticals where stable relations among SSC, discharge, and sensor signal have been established. Multi-vertical sampling or moving ADCP surveys should be used regularly to check lateral representativeness. If the objective is spatial distribution in estuaries, lakes, reservoirs, or large river reaches, fixed stations, mobile profiles, automatic sampling, and remote sensing should be integrated.

The workflow can be summarized as follows. First, determine whether the cross section is stable, mixing is sufficient, and SSC lies within the effective instrument range. Second, choose single-point, two-point, multi-vertical, or mobile survey schemes according to the lateral gradient. Third, choose fixed-depth, multi-point vertical, pumped-profile, or acoustic-profile schemes according to the vertical gradient. Finally, verify conversion relations using isokinetic field sampling, automatic sampling, and historical data, and include vertical representativeness error, instrument error, and calibration error in the uncertainty assessment. This framework turns the question of whether one or two verticals are enough into a verifiable conditional judgment rather than an empirical conclusion.

To clarify the scale conversion from fixed-point online measurements to sediment-load indices, Figure 4 conceptualizes the estimation process from point measurement to single-vertical and cross-sectional sediment load. A point sensor first obtains local SSC and velocity at its installed depth, namely, $C_p(t)$ and $u_p(t)$. With vertical distribution information, multi-point comparison, or empirical weights, the suspended-sediment flux of that vertical, $q_{s,v}(t)$, can be estimated. Only after lateral representativeness, vertical control width, and cross-sectional discharge distribution

are considered can it be expanded to cross-sectional sediment load, $Q_s(t)$. A point online measurement is therefore not directly equivalent to cross-sectional sediment load and requires scale conversion and representativeness correction from point to vertical and from vertical to section.

Based on cross-section stability, lateral and vertical representativeness, monitoring objectives, and field calibration conditions, Figure 5 further summarizes the deployment decision framework. For trend warning, fixed OBS or ABS sensors combined with event sampling can be prioritized. For cross-sectional sediment load, the coupling of representative verticals, acoustic profiles, and discharge data is central. For spatial distribution at reach, lake-reservoir, or estuarine scales, fixed stations, moving profiles, and remote sensing information should be integrated. Regardless of the scheme used, sampling design, calibration relations, representativeness error, quality control, and uncertainty ranges must be reported.

4 Sediment *in situ* monitoring accuracy analysis

4.1 Environmental factors affecting *in situ* online measurement

In situ measurement of SSC in flowing water depends strongly on instrument sensitivity, resolution, stability, and repeatability (Sa'ad et al., 2021). Other studies summarize the main controls on accuracy as instrument performance, sediment-transport characteristics, and environmental interference (Fettweis et al., 2019).

Instrument sensitivity and resolution are the basic indicators that determine measurement accuracy. The performance of sensing elements and the response relations between acoustic or optical signals and SSC directly determine the ability and stability of an instrument in identifying concentration changes. During long-term operation, calibration state, electrical-signal stability, and contamination of observation windows also accumulate effects on measurement results. Contact instruments deployed in water may also disturb the local flow field and introduce additional error.

Suspended sediment transport in flowing water is strongly turbulent. Velocity fluctuation, turbulent structures, bed-sediment resuspension, and flood rise and recession cause multi-scale temporal and spatial fluctuations in SSC. This heterogeneity is especially pronounced during floods or under high-concentration conditions. Acoustic and optical signals propagate rapidly and sensors can have high response frequencies, but this does not mean that measurement duration and velocity effects can be ignored. Sampling duration, averaging window, velocity fluctuation, and turbulence scale directly affect the representativeness of point SSC estimates and can cause Type I errors related to insufficient sampling time. During floods, rapid changes in SSC and particle-size composition may also move actual concentrations beyond the effective instrument range or existing calibration relation, increasing random error and systematic bias at the same time.

Interference in complex aquatic environments also cannot be ignored. Aquatic vegetation, plankton, bubbles, and other

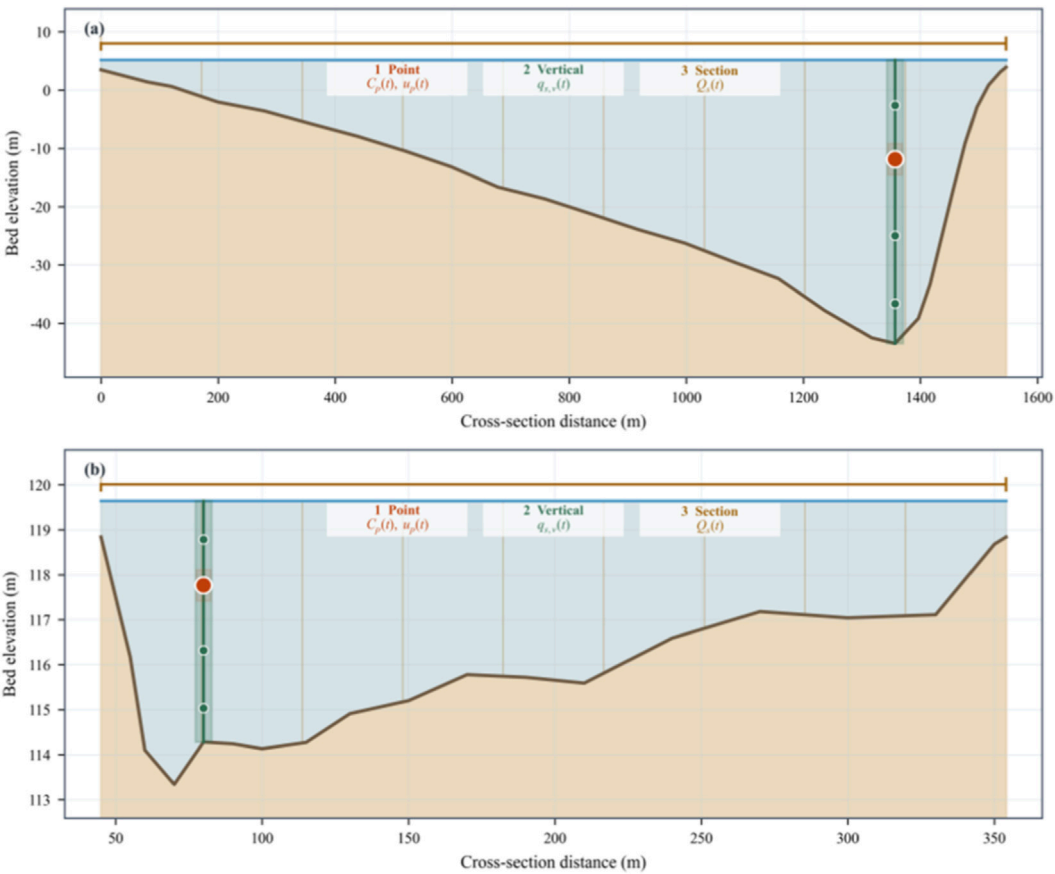


FIGURE 4
Conceptual upscaling from point SSC and velocity measurements to vertical and cross-sectional suspended-sediment flux. (a) Yangtze river, Nanjing station, 2023. (b) Yellow river, Xixiayuan station, 2025.

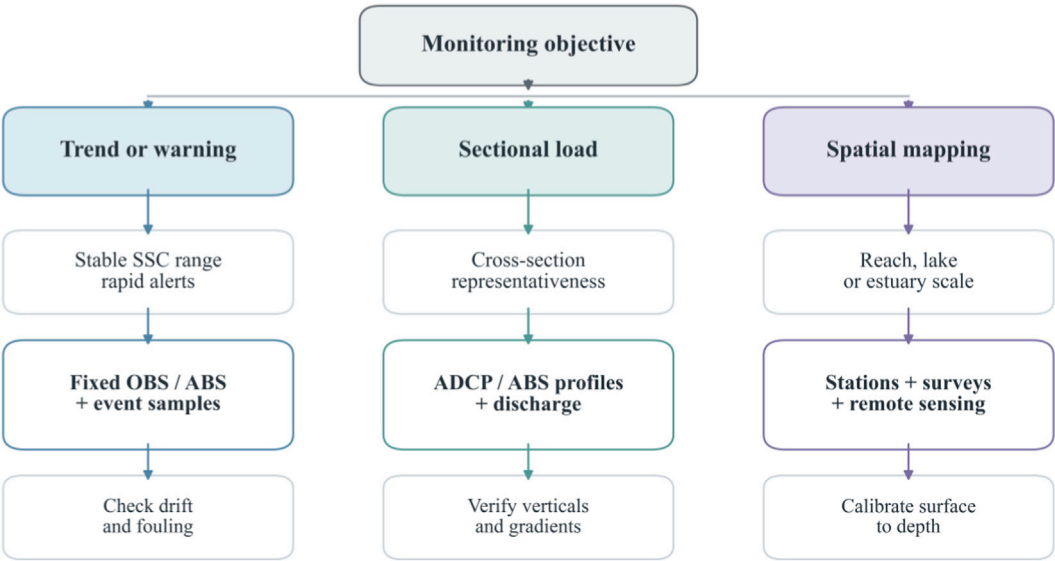


FIGURE 5
Decision framework for selecting suspended sediment concentration (SSC) monitoring configurations.

constituents can alter medium properties and interfere with the propagation path and attenuation of acoustic or optical signals, reducing measurement stability and reliability. Water temperature changes can further increase uncertainty by altering physical water properties and sensor operating states. The coupled effects of these factors under different hydrological conditions make accurate SSC measurement challenging.

4.2 Main influencing factors in laboratory calibration

Calibration under any condition requires a quantitative relation between SSC and sensor signal. OBS calibration experiments in the turbidity maximum zone of the Yangtze Estuary show substantial error because laboratory calibration cannot fully reproduce field conditions (Liu et al., 2006). In laboratory experiments for high-concentration suspended sediment, stirring or circulation systems are usually added to reduce nonuniform particle-size distribution and the effects of salinity and fine-particle flocculation (Zhang et al., 2019). Laboratory calibration is often conducted in still water or idealized controlled flow and therefore tends to neglect hydrodynamic effects. Field measurements, however, are strongly affected by velocity variation, which changes suspended particle-size distribution (Schoellhamer and Wright, 2003) and acoustic attenuation characteristics (Sassi et al., 2012), causing calibration parameters obtained in the laboratory to deviate in complex natural dynamic flows.

For low-concentration calibration with $SSC < 10 \text{ kg/m}^3$, a linear relation with $R^2 > 0.99$ is commonly obtained, whereas high-concentration calibration is generally nonlinear. The OBS-3A output enters a saturation zone at $10\text{--}30 \text{ kg/m}^3$ and then decreases as concentration increases. Below 10 kg/m^3 , the relation is linear; above this range, power-function fitting is often used. Because the difference from low-concentration calibration is large, many studies use segmented calibration at high concentration (Wang et al., 2020; Liu et al., 2006). The maximum-response turning point of sensor calibration also differs among high-concentration laboratory experiments, so special attention should be paid to the turning point when calibrating for high SSC (Zhang et al., 2019). Differential-pressure methods can avoid optical-sensor saturation by retrieving concentration from particle settling or density-related effects, but they still require calibration with physical samples (Fettweis et al., 2019).

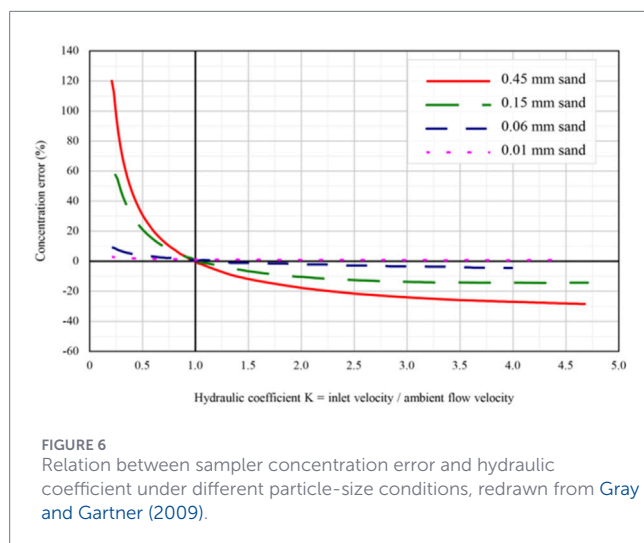
Calibration accuracy directly determines the reliability of online monitoring. Instantaneous samplers can produce large point-scale random errors because of sediment fluctuation, and repeated sampling, time-integrated sampling, or combined use with automatic samplers is needed to reduce random bias (Fettweis et al., 2019). Long-term operation should also include blank samples, comparisons before and after cleaning, cross-sensor checks, intensified event sampling, sensor zero and span checks, and residual monitoring to identify biofouling, window blockage, electronic drift, acoustic transducer aging, and changes in mounting attitude. Recent USGS work on submersible acoustic backscatter sensors shows that regular calibration checks and long-term drift assessment should be part of operational SSC monitoring rather than optional steps in data post-processing (Alexander et al., 2023).

4.3 Effects of field samplers in paired measurements

Before online methods are used to determine SSC, calibration and comparison are required. In addition to laboratory calibration using prepared water samples with known SSC, field comparison generally uses sampling, filtration, and oven drying. The selection and use of sampling instruments are therefore critical to the accuracy of field comparison. Sampler development has a history of more than 200 years, and during the past century much work has focused on finding an ideal sampler. Nearly 100 sampler types are currently available worldwide (Haddadchi, 2017; Qiu, 1996; U.S. Geological Survey, 2026).

According to time control, samplers can be divided into instantaneous samplers and time-integrated samplers. Instantaneous samplers collect water at a specific moment and are easy to operate, but they are strongly affected by sediment fluctuation and the collected sample can have poor representativeness. Time-integrated samplers include point-integrating and depth-integrating samplers. Point-integrating samplers collect water at a fixed point over a defined period, whereas depth-integrating samplers collect an integrated sample along the vertical. Compared with instantaneous samplers, time-integrated samplers collect a composite sample over a time interval and therefore have stronger representativeness, but their temporal resolution is lower and they cannot satisfy high-frequency monitoring needs (Gray and Gartner, 2009). In terms of structure, samplers include bottle, vertical tubular, pressure-regulated chamber, and bag types (Qiu, 1996). Small samplers are often handheld for flexible field operation, whereas large time-integrated samplers require cableways, boats, bridge inspection vehicles, or similar platforms for field measurement.

According to whether the intake velocity matches ambient flow velocity, samplers can be divided into non-isokinetic and isokinetic types. Most instantaneous samplers and conventional self-developed open samplers are non-isokinetic, because their intake velocity cannot match the natural flow velocity. Time-integrated samplers can adjust intake velocity through regulating devices so that it matches the surrounding flow, and they are increasingly used at hydrological stations (Guy, 1970). The US D-99 depth-integrating sampler can operate to 67 m depth and is suitable for deep-water measurement (Rubin and Wright, 2013). Time-integrated samplers also have minimum applicable velocity requirements. Rigid time-integrating samplers require a sampling-flow velocity of at least 0.46 m/s , and bag-type time-integrating samplers require at least 0.61 m/s (Haddadchi, 2017). Intake velocity is mainly adjusted through coordinated valve and exhaust devices. Previous work confirms that mismatch between sampler intake velocity and ambient flow velocity can introduce large SSC measurement errors. The hydraulic coefficient K is commonly used as the ratio of intake velocity to natural flow velocity. When $K > 1$, measured SSC becomes progressively lower than the true concentration as K increases. When $K < 1$, measured SSC becomes progressively higher than the true concentration as K decreases. The bias increases with particle size and can reach 130% (Figure 6). For time-integrated samplers, the industry commonly uses $K = 1$ or variation within $\pm 10\%$ as the criterion for applicability. Field comparisons show, however, that this criterion applies only to conventional conditions with SSC below 100 kg/m^3 , and sampling under high-concentration conditions still



requires further study. Lower temperature increases water viscosity and reduces K (Rubin and Wright, 2013). Because particle size also matters, field testing and calibration should be strengthened under low-temperature conditions, especially when coarse sand is sampled.

The hydraulic coefficient K is the ratio of sampler intake velocity to natural flow velocity. The figure illustrates the direction and magnitude of non-isokinetic sampling error with changes in K and particle size. Specific values require recalibration according to sampler structure, flow velocity, temperature, and particle-size distribution.

4.4 Accuracy assessment and error composition of *in situ* sediment monitoring

Errors in online sediment monitoring are jointly controlled by instrument characteristics, environmental disturbance, and sampling and deployment strategies (Thorne and Hanes, 2002; Downing, 2006). Statistically, measurement errors can be divided into random errors, systematic errors, and gross errors that should be removed through quality control (Horsburgh et al., 2015). Given the unsteady nature of suspended sediment transport and its strongly three-dimensional spatial distribution, online monitoring errors can be further grouped into instrument measurement errors and spatial-temporal sampling errors (Horowitz, 2003).

Error sources differ across spatial scales. At the point scale, errors mainly come from instrument resolution and response characteristics, as well as Type I errors caused by insufficient sampling duration to represent sediment fluctuation. At the vertical scale, when vertical mean SSC is calculated, limited point number and approximate integration methods introduce Type II errors in addition to the effects of instrument weighting and point-position drift. At the cross-sectional scale, vertical density and spatial arrangement directly affect representativeness and accuracy in sediment-load estimation. Inadequate deployment or insufficient survey-line density leads to incomplete spatial information and produces cross-sectional, or Type III, errors (Niu, 1999). The consistency and applicability of instrument calibration, the accuracy of *in situ* sampling, and the match between sampling and online measurements also strongly affect error accumulation and result reliability.

Accuracy evaluation should use uncertainty to characterize the dispersion and confidence interval of measurement results, rather than reporting only a single relative error (Pelletier, 1988). For complex hydrological cross sections, uncertainty in sediment-load estimation follows error-propagation principles and consists of both random and systematic uncertainty (Le Coz et al., 2012; Harmel et al., 2006). The combination is nonlinear. Random uncertainty decreases as sampling density and averaging time increase, and its estimation should include contributions from Type I, Type II, and Type III random errors and instrument random uncertainty. Systematic uncertainty has a spatial accumulation effect and appears as the superposition of systematic bias across scales. It is closely related to instrument calibration, deployment design, *in situ* verification, data cleaning, and long-term drift control.

4.5 Automated quality control, drift correction, and uncertainty reporting

The high temporal resolution of online monitoring has hydrological value only when supported by a QA/QC system. Continuous SSC records should preserve raw signals, cleaning and maintenance logs, instrument status, anomaly-removal rules, time differences between sampling and sensor observations, regression-model versions, and uncertainty-estimation methods. Sudden jumps, flat-line behavior, sensor saturation, zero drift, window contamination, abnormal acoustic echoes, bubble interference, and communication gaps should be flagged through automated rules combined with manual review. Experience from continuous water-quality sensors shows that open-source or standardized data-review workflows help improve the consistency and traceability of long-term records (Horsburgh et al., 2015; Wagner et al., 2006).

Result reports should provide at least four types of information. The first is the calibration model from sensor signal to SSC, including sample number, covered concentration and particle-size ranges, residual statistics, and extrapolation boundaries. The second is the field comparison method, including sampler type, isokinetic status, sampling duration, spatial matching, and laboratory quality control. The third is the data-processing workflow, including anomaly removal, averaging window, drift correction, gap filling, and version records. The fourth is the source and synthesis method for SSC and sediment-load uncertainty. Only when instrument performance, calibration residuals, spatial-temporal representativeness, and cross-sectional integration error are reported together can online sediment-monitoring results support cross-site comparison and model assimilation.

5 Conclusion

This technical review summarizes the principles, applicability, deployment modes, and accuracy-control mechanisms of online suspended sediment monitoring technologies. It compares the advantages and limitations of mainstream optical and acoustic technologies under different water-sediment conditions and analyzes uncertainty sources related to instrument characteristics, environmental disturbance, calibration, field sampling, quality control, and cross-sectional representativeness. The review shows that the core challenge in online suspended sediment monitoring is not merely sensor sensitivity, but the coupled problem of converting

surrogate signals into SSC, extending point observations to vertical and cross-sectional scales, and controlling drift and systematic bias during long-term operation. Given this state of the field, future research should focus on the following directions.

Acoustic and optical methods should be integrated by combining the high temporal resolution and low cost of optical technologies with the profiling capability and low-light adaptability of acoustic technologies. Multi-frequency, multispectral, synchronous particle-size observation, and Bayesian or machine-learning retrieval frameworks should be introduced to reduce particle-size dependence, high-concentration signal saturation, and the risk of single-sensor failure.

Quantitative research on vertical and point deployment should be strengthened, and deployment decision frameworks should be developed according to monitoring objectives and uncertainty thresholds. The number of representative verticals, fixed-point depth, and mobile-survey frequency should be determined by cross-section stability, lateral and vertical SSC gradients, mixing state, and historical comparison data, rather than by directly applying experience from a small number of stations.

A combined laboratory and field calibration system is needed. Laboratory calibration should better simulate natural water-sediment conditions, particle-size distribution, flocculation, salinity, and velocity variation, while field isokinetic sampling, automatic sampling, segmented calibration, calibration checks, and drift correction should improve adaptability under both high and low concentrations.

Instrument development, deployment strategy, calibration, automated QA/QC, and uncertainty reporting should be integrated into a reproducible data-processing chain from the perspective of the entire observation workflow. For SSC monitoring that supports ecological and biogeochemical research, coordinated observations of particulate carbon, nitrogen, phosphorus, contaminant fluxes, light environment, primary productivity, algal bloom indicators, estuarine turbidity maxima, and habitat indicators should also be strengthened. Such paired monitoring would make online SSC networks useful not only for sediment-load estimation but also for carbon and nutrient budgets, sediment-associated pollutant-flux assessment, light-limitation diagnosis, bloom early warning, and estuarine habitat evaluation.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding authors.

Author contributions

FZ: Writing – original draft, Formal Analysis, Project administration, Visualization, Resources, Data curation, Validation, Methodology, Supervision, Software, Investigation, Writing – review and editing, Conceptualization, Funding acquisition. QY: Methodology, Data curation, Project administration, Investigation, Validation, Software, Writing – original draft, Supervision, Writing – review and editing, Formal Analysis, Conceptualization. ZY: Data

curation, Conceptualization, Writing – original draft, Formal Analysis, Writing – review and editing. YW: Writing – review and editing, Data curation, Conceptualization, Formal Analysis. JL: Data curation, Conceptualization, Formal Analysis, Writing – review and editing. HX: Formal Analysis, Conceptualization, Data curation, Writing – review and editing. MX: Methodology, Writing – review and editing, Project administration, Investigation. YW: Writing – review and editing, Project administration, Methodology, Investigation, Funding acquisition. RT: Methodology, Data curation, Software, Writing – review and editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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